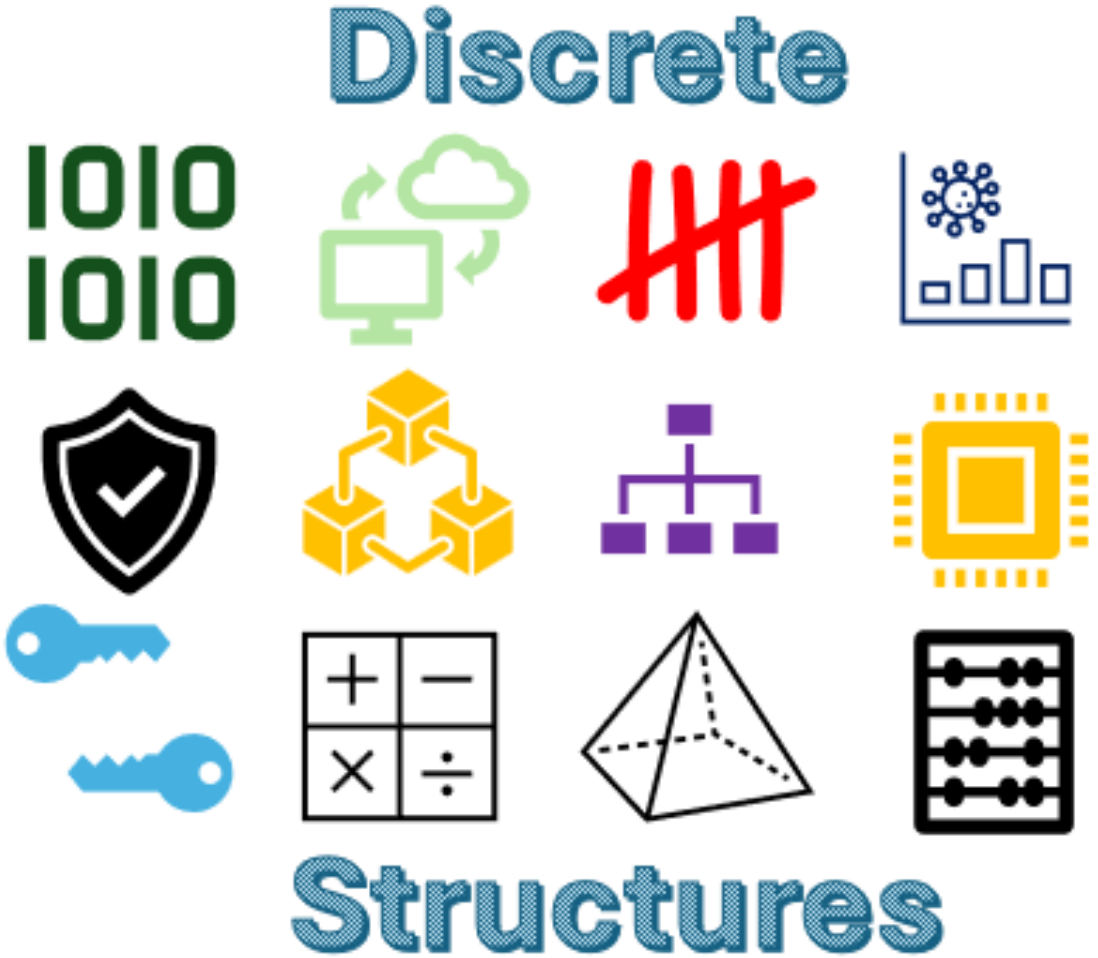
COMP 2310: Discrete Structures
Fall 2026

Chapter 4: Relation and digraph



Transitive Closure and Warshall's Algorithm

Transitive closure

- If R is a relation on a set A, the transitive closure R^{∞} contains all pairs (a, b) where there's a path in R from a to b.
- Identifies reachability in directed graphs.
- Useful in many applications such as: pathfinding and network analysis.

Transitive closure: Example

- Given a set A and relation R:
  - A={1,2,3,4}
  - R={(1,2),(2,3),(3,4),(2,1)}
  - Find R^{∞} = ?
    - Determine all reachable pairs in R to construct R^{∞}
    - R^{∞} = {*(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4),* (3, 4)}

How to find transitive closure

- Given a set A and relation R:
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- **Approaches to find transitive closure:**
  - Graph approach
  - Matrix approach

How to find transitive closure: graph approach

- A={1,2,3,4}
- R={(1,2),(2,3),(3,4),(2,1)}
- Graph approach
  1. Draw a digraph
  2. Find all possible pairs
     - R^{∞} = {*(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4)*}

How to find transitive closure: matrix approach

- A={1,2,3,4}
- R={(1,2),(2,3),(3,4),(2,1)}
- Matrix approach
  1. Construct the adjacency matrix, M_R
  2. Multiply M_R up to the number of elements in A (e.g. up to M^4_R for the example)
  3. Perform OR (Union) operations with all powers of M_R (up to the number of elements in A)
  4. Construct the R^∞ from the resulting matrix
     - R^∞ = {*(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4)*}

Challenges in finding transitive closure

- Graph method is impractical for larger graph
- Matrix method is inefficient due to large number of matrix multiplications

Efficient approach to finding transitive closure

Warshall's algorithm

Warshall's algorithm for finding transitive closure

- Compute the transitive closures of a relation using adjacency matrices
- Iterative approach to make connections between vertices
- **Step 1:** transform the relations into the adjacency matrix, M_R
- **Step 2:** initialize $W_0 = M_R$
- **Step 3:** Iterate for k = 1, 2, …, n (n is the number of elements in the set or num_rows or num_column in the adjacency matrix):

  3.1. For W_k list the locations $p_1, p_2, \ldots$, in column k of $\mathbf{W}_{k-1}$, where the entry is 1, and the locations $q_1, q_2, \ldots$, in row k of $\mathbf{W}_{k-1}$, where the entry is 1.

  3.2. Put 1's in all the positions p_i, q_j of $\mathbf{W}_k$ (if they are not already there)

- W_n (the last matrix) represents the transitive closure

Warshall's algorithm for finding transitive closure: example

- A = {1, 2, 3, 4}
- R = {(1,2), (2,3), (3,4), (2,1)}

Steps 1 and 2: $W_0 = M_R$

	1	**2**	**3**	**4**
1	0	1	0	0
2	1	0	1	0
3	0	0	0	1
4	0	0	0	0

- Row i, column j is 1 when (i, j) is in R
- Example: (2, 1) is in R, so row 2, column 1 is 1
- Next: repeat for k = 1, 2, 3, 4 to build W_1, W_2, W_3 and W_4

Warshall's algorithm: k = 1

W_0 (look at column 1 and row 1)

	1	2	3	4
1	0	**1**	0	0
2	**1**	0	1	0
3	0	0	0	1
4	0	0	0	0

Put 1's at (p, q)



W_1

	1	2	3	4
1	0	1	0	0
2	1	**1**	1	0
3	0	0	0	1
4	0	0	0	0

Blue: column 1 and row 1 of W_0 **Orange:** new 1's added in W_1

- Column 1 of W_0 has a 1 in row 2, so p = { 2 }
- Row 1 of W_0 has a 1 in column 2, so q = { 2 }
- Put a 1 at (p, q) = (2, 2). It is not there yet, so it is the only change in W_1

Warshall's algorithm: k = 2

W_1 (look at column 2 and row 2)

	1	2	3	4
1	0	**1**	0	0
2	**1**	**1**	**1**	0
3	0	0	0	1
4	0	0	0	0

Put 1's at (p, q)



W_2

	1	2	3	4
1	**1**	1	**1**	0
2	1	1	1	0
3	0	0	0	1
4	0	0	0	0

Blue: column 2 and row 2 of W_1 **Orange:** new 1's added in W_2

- Column 2 of W_1 has 1's in rows 1 and 2, so p = { 1, 2 }
- Row 2 of W_1 has 1's in columns 1, 2 and 3, so q = { 1, 2, 3 }
- Put 1's at all (p, q): (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3). New: (1, 1), (1, 3). The other four were already 1

Warshall's algorithm: k = 3

W_2 (look at column 3 and row 3)

	1	2	3	4
1	1	1	**1**	0
2	1	1	**1**	0
3	0	0	0	**1**
4	0	0	0	0

Put 1's at (p, q)



W_3

	1	2	3	4
1	1	1	1	**1**
2	1	1	1	**1**
3	0	0	0	1
4	0	0	0	0

Blue: column 3 and row 3 of W_2 **Orange:** new 1's added in W_3

- Column 3 of W_2 has 1's in rows 1 and 2, so p = { 1, 2 }
- Row 3 of W_2 has a 1 in column 4, so q = { 4 }
- Put 1's at (1, 4), (2, 4). Both are new

Warshall's algorithm: k = 4 and the result

W_3 (look at column 4 and row 4)

	1	2	3	4
1	1	1	1	**1**
2	1	1	1	**1**
3	0	0	0	**1**
4	0	0	0	0

Put 1's at (p, q)



W_4

	1	2	3	4
1	1	1	1	1
2	1	1	1	1
3	0	0	0	1
4	0	0	0	0

Blue: column 4 and row 4 of W_3 **Orange:** new 1's added in W_4

- Column 4 of W_3 has 1's in rows 1, 2 and 3, so p = { 1, 2, 3 }
- Row 4 of W_3 is all 0, so q = { } and nothing is added. $W_4 = W_3$
- W_4 is the transitive closure: R^{∞} = {(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,4)}

Exercise

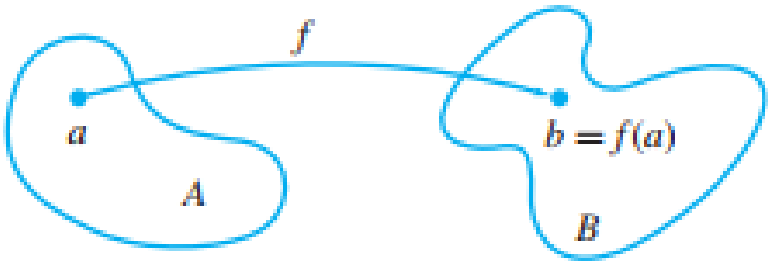
- Find the transitive closure using Warshall's algorithm for the following matrix:

$$\mathbf{M}_R = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Chapter 5: Functions

Function

- Function is a special type of relation
- A function f:A→B is a relation from set A to set B
  - A is the domain of the function f, which represents all the possible inputs that the function can take.
  - B is the codomain, which represents the set of all possible outputs that the function can produce
  - For each a∈Dom(f) there is exactly one element f(a)∈B
- If a∉Dom, then f(a)=∅
- f(a)=b where b is the image of a



Function: example

- Let $A = \{1, 2, 3, 4\}$
- $B = \{a, b, c, d\}$, and let
- $f = \{(1,a), (2,a), (3, d), (4, c)\}$.
  - $f(1) = a$
    $f(2) = a$
    $f(3) = d$
    $f(4) = c$

Function: example

- Let $A = \{1, 2, 3, 4\}$
- $B = \{x, y, z\}$
- $R = \{(1, x), (2, x)\}$
- $S = \{(1, x), (1, y), (2, z), (3, y)\}$
- S is not a function since, f(1) = {x, y}

Function: example

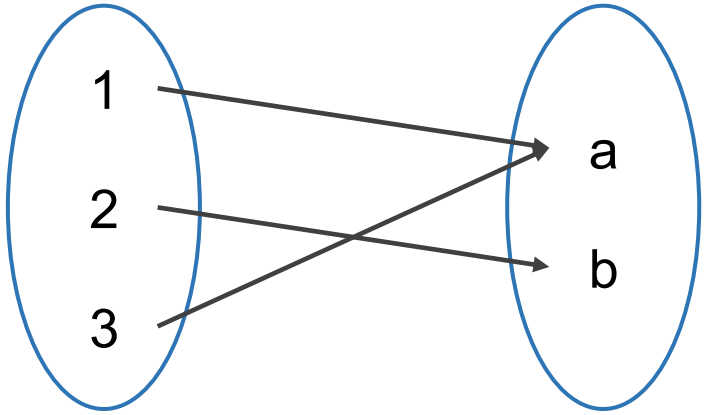
- Let $A = \mathbb{Z}$ and let $B = \{0, 1\}$. Let $f : A \to B$ be found by

$$f(a) = \begin{cases} 0 & \text{if } a \text{ is even} \\ 1 & \text{if } a \text{ is odd.} \end{cases}$$

Function or not a function?

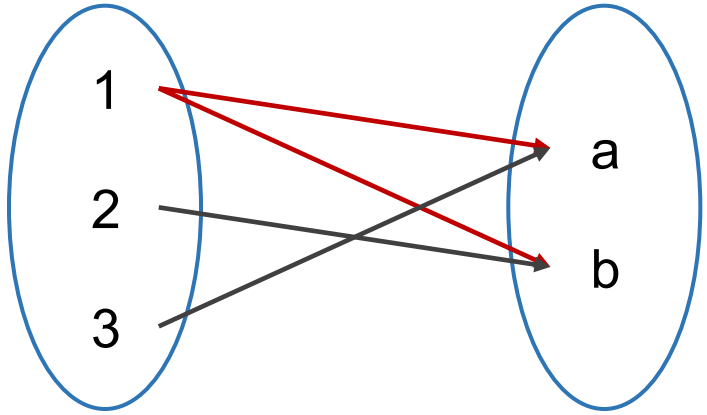
- Test: every element of A is sent to exactly one element of B. Two inputs may share the same output.

$f = \{(1, a), (2, b), (3, a)\}$



Function: each input has exactly one output

$g = \{(1, a), (1, b), (2, b), (3, a)\}$



Not a function: 1 is sent to both a and b

- $f(x) = x^2$ on the integers is a function: $f(2) = f(-2) = 4$ is allowed
- $y^2 = x$ is not a function: $x = 4$ gives both $y = 2$ and $y = -2$

Identity function

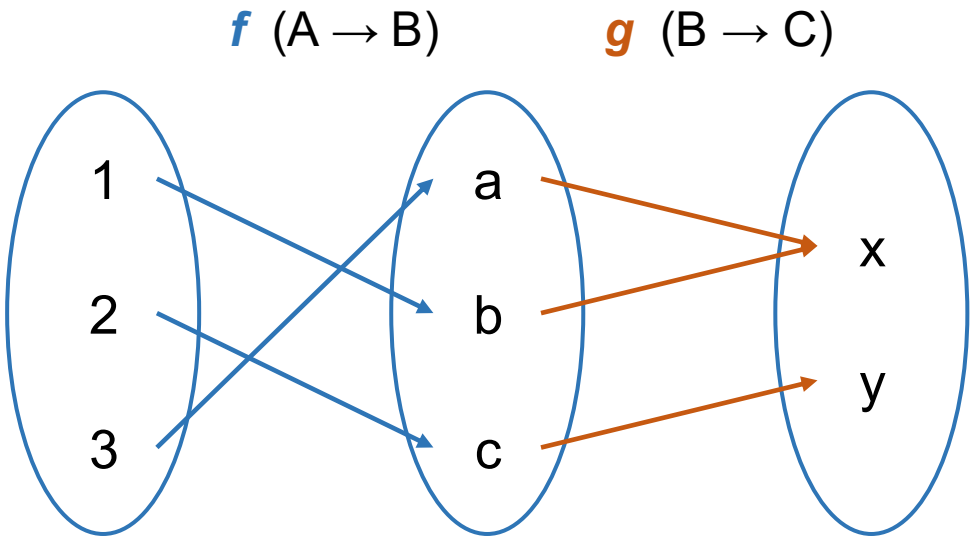
- Let A be a nonempty set
- The identity function on A, denoted by I_A is defined as:
  - $I(a)_A = a$
  - Let A={1,2,3}
  - The identity function I_A can be represented as:
    - $I_A(1)=1$
    - $I_A(2)=2$
    - $I_A(3)=3$

Function composition

- Given function f: A→B and g: B→C, the composition f and g is defined by $g \circ f$ is defined by:
  - $(g \circ f)(a) = g(f(a))$
    - Apply f to a to get f(a)
    - Then apply g to f(a) to get g(f(a))
  - Example:
    - f(a)=a+2
    - g(b) = 3b
    - Then the composition $g \circ f$ *is:*
      - $(g \circ f)(a) = g(f(a)) = g(a+2) = 3(a+2) = 3a + 6$

Function composition: more examples

- $f(1) = b$, $f(2) = c$, $f(3) = a$
- $g(a) = x$, $g(b) = x$, $g(c) = y$
- $(g \circ f)(1) = g(f(1)) = g(b) = x$
- $(g \circ f)(2) = g(f(2)) = g(c) = y$
- $(g \circ f)(3) = g(f(3)) = g(a) = x$
- $g \circ f = \{(1, x), (2, y), (3, x)\}$



- **Order matters.** Take $f(a) = a + 2$ and $g(b) = 3b$
- $(g \circ f)(a) = g(a + 2) = 3a + 6$, but $(f \circ g)(a) = f(3a) = 3a + 2$
- For $a = 1$: $(g \circ f)(1) = 9$ and $(f \circ g)(1) = 5$, so $g \circ f \neq f \circ g$